

COURSE DESCRIPTION			
Program	Polymer Technology, Printing Technology, Textile Technology, Chemical Engineering		
Course Title	Chemistry for Technical Practices		
Course Code	2003B	Semester	II
Type of Course	Practicum	Contact Hours	90
Teaching Scheme (L: T:P:R)	3:0:3:0	Credits	4.5
CIE Marks	40	ESE Marks	60

Introduction

Chemistry for technical practices course introduces essential topics in Chemistry for scientific foundation and also emphasizes the development of scientific thinking through laboratory experiences which help students connect theoretical concepts with real industrial and domestic applications.

Course Objectives

1	To impart knowledge about fundamental concepts in chemistry including atoms, molecules, solutions, thermodynamics and relate them to engineering applications.
2	To apply key concepts of analytical chemistry and water treatment methods for solving real-world domestic and industrial problems.
3	To impart basic knowledge about organic functional groups, dyes and pigments
4	To familiarize students with various engineering materials for selecting appropriate materials in industrial and domestic applications and understand green chemistry principles.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge of Chemistry			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply the basic concepts of atoms, molecules, solutions and thermodynamic processes in scientific and engineering contexts.	Apply
CO2	Apply the fundamentals of analytical Chemistry, appropriate water treatment methods and also the basic concepts of acids and bases to solve engineering problems.	Apply
CO3	Examine the structure, classification, and applications of dyes, pigments and organic functional groups through theoretical and experimental approaches.	Apply
CO4	Explore various engineering materials for suitable industrial and domestic applications and explain green Chemistry principles promoting sustainable practices.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	-
CO2	3	3	2	3	2	2	2	-	-	-	-
CO3	2	2	2	2	2	-	-	-	-	-	-
CO4	3	2	3	2	2	2	3	-	-	-	-
Total	11	9	7	7	6	4	5	-	-	-	-
Correlation Level	2.75	2.25	2.33	2.33	2	2	2.5	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
3	0	3	0	6	90	4.5	20	60	80	20	0	20	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Fundamentals of Chemistry, Basic concepts of solution and Thermodynamics	Atoms and molecules: Fundamental particles- electrons, protons and neutrons, their charge and mass - Atomic number and mass number. Solutions: Solute, solvent and standard solution- methods to express the concentration of solution:- molarity, normality and ppm -Problems based on molarity and normality. Thermodynamics: System and surroundings - State variables (T, P &V)- Open, closed and isolated systems - Isothermal, isobaric, isochoric, adiabatic, reversible, irreversible, spontaneous, nonspontaneous, endothermic and exothermic processes - Internal energy, enthalpy and entropy.	11	0
Module I	Experiments on preparation of standard solutions & familiarization of endothermic and exothermic reactions	Preparation of 0.1 N oxalic acid solution, 0.1 N Sodium Carbonate solution, 0.1 M Zinc Sulphate solution. 0.1 M Magnesium Sulphate solution and demonstration of endothermic and exothermic reactions.	0	9
Module II	Basic concepts of Analytical Chemistry, Water treatment methods and key concepts of acids and bases	Volumetric Analysis: Titration, end point, indicators and standard solution, choice of indicators in acid-base titration- Principle of Volumetric Analysis ($V_1N_1=V_2N_2$), problems. Water: Soft and hard water - Salts causing water hardness - Temporary and permanent hardness. Water Treatment: Effluent treatment methods-Physical, chemical, biological and advanced technique (Flow chart) - Potable water and its characteristics- Municipal water treatment methods (in brief only): Screening, sedimentation, coagulation, filtration and sterilization:- Chlorination, UV radiation, and ozonation- Flow chart for municipal water treatment-Water filtration methods: Reverse osmosis and nanofiltration. Acids and bases (Arrhenius concept)- pH and pOH- pH equation- pH scale- Relation between pH and pOH - Simple problems based on pH - Applications of pH.	12	0
Module II	Quantitative & Qualitative Analysis	Standardization of HCl using standard sodium carbonate solution and standardization of EDTA using $ZnSO_4$. Estimation of sodium hydroxide using standard hydrochloric acid, potassium hydroxide solution using standard oxalic acid solution, total hardness of given water sample using standard EDTA solution. pH Determination using different techniques- pH meter, Universal indicator and pH test paper.	0	12
Module III	Organic Compounds with some important Functional Groups, Dyes and Pigments	Organic compounds - Functional group (alcohol, acid, ester, hydrocarbon) - Basic concept Dyes - definition, basic concept of chromophore, auxochrome-examples, classification by- source, application method, chemical structure. Dye-fibre interactions (physical adsorption, hydrogen bonding, ionic & covalent interactions) - basic concept. Pigments - types (inorganic and organic) and uses.	11	0
Module III	Experiments on preparation of dyes, esterification, identification of functional groups and test for unsaturation, and aromaticity	Preparation of dyes Esterification Identification of functional groups Test of saturated and unsaturated organic compounds Test for aromaticity	0	9

Module IV	Engineering Materials & Green Chemistry	Alloys: Purposes of alloying, applications of alloys like stainless steel, brass, bronze, and solder. Polymers: Monomers, polymers, polymerization, homo and copolymers, thermoplastics and thermosetting plastics- common polymers: PE, PVC, Nylon-66, Bakelite, PET, PU, PBT, PC, and PMMA- Applications of each polymer- Natural rubber, vulcanization of rubber, and properties of vulcanized rubber- Synthetic rubber. Nanomaterials and nanotechnology- Carbon nanotubes, fullerenes, graphene- Applications of nano materials. Metal Organic Frameworks (MOFs) as advanced engineering materials - Applications. Biomaterials: Metallic, ceramic, polymeric and composites - Applications of biomaterials. Biofuel and its applications. Green Chemistry - Principles, importance and applications.	11	0
Module IV	Experiments based on estimation of metals in alloys and preparation of polymers	Estimation of metals in alloys and preparation of polymers.	0	9

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Estimation of metals in alloys and preparation of polymers.	Define atom and molecule with examples.	L	CO1	Remember	0.5
Estimation of metals in alloys and preparation of polymers.	List the fundamental particles of atom (electrons, protons and neutrons) and give their charge and mass.	L	CO1	Understand	0.5
Estimation of metals in alloys and preparation of polymers.	Define the terms atomic number and mass number with examples.	L	CO1	Remember	0.5
Estimation of metals in alloys and preparation of polymers.	Calculate the number of fundamental particles of an atom using atomic number and mass number of elements (upto atomic number Z=20).	L	CO1	Apply	0.5
Solutions	Define the terms solute, solvent and standard solution with examples.	L	CO1	Remember	1
Solutions	Define the terms molarity, normality and ppm and give the equations for molarity, normality and ppm.	L	CO1	Understand	1
Solutions	Solve problems based on molarity and normality.	L	CO1	Apply	2
Thermodynamics	Define the terms system and surroundings in thermodynamics and the state variables (T, P & V)	L	CO1	Remember	1
Thermodynamics	Explain different types of systems- open, closed and isolated systems with example.	L	CO1	Understand	1
Thermodynamics	Explain thermodynamic processes- isothermal, isobaric, isochoric and adiabatic with example.	L	CO1	Understand	0.5
Thermodynamics	Differentiate between reversible and irreversible processes with example.	L	CO1	Understand	0.5
Thermodynamics	Differentiate between spontaneous and non spontaneous processes with example.	L	CO1	Understand	0.5
Thermodynamics	Differentiate between endothermic and exothermic processes with example.	L	CO1	Understand	0.5
Thermodynamics	Define the terms internal energy, enthalpy and entropy.	L	CO1	Remember	1
Preparation of standard solutions and demonstration of endothermic and exothermic reactions.	Prepare 0.1 N oxalic acid solution	P	CO1	Apply	1.5
Preparation of standard solutions and demonstration of endothermic and exothermic reactions.	Prepare 0.1 N sodium carbonate solution	P	CO1	Apply	1.5

Preparation of standard solutions and demonstration of endothermic and exothermic reactions.	Prepare 0.1 M Zinc Sulphate solution	P	CO1	Apply	1.5
Preparation of standard solutions and demonstration of endothermic and exothermic reactions.	Prepare 0.1 M Magnesium Sulphate solution	P	CO1	Apply	1.5
Preparation of standard solutions and demonstration of endothermic and exothermic reactions.	Demonstrate endothermic and exothermic reactions (Endothermic: The reaction of citric acid and baking soda, the reaction of barium hydroxide and ammonium chloride; Exothermic: Neutralisation between acids and bases- (e.g., HCl and NaOH reaction,) and reaction of NaOH in water, reaction of CaCl ₂ in water	P	CO1	Apply	3
Module II					
Volumetric analysis	Define the terms titration, end point, indicator and standard solution in volumetric analysis.	L	CO2	Remember	0.5
Volumetric analysis	Explain the choice of indicator in the following titrations: Strong acid- strong base, strong acid-weak base and weak acid-strong base (with example).	L	CO2	Understand	0.5
Volumetric analysis	State the Principle of Volumetric Analysis using normality equation ($V_1N_1 = V_2N_2$).	L	CO2	Remember	0.5
Volumetric analysis	Solve problems based on normality equation $V_1N_1 = V_2N_2$.	L	CO2	Apply	1
Hardness of water	Differentiate between soft and hard water. List the salts causing hardness of water. Differentiate between temporary and permanent hardness of water.	L	CO2	Understand	1
Water treatment methods	Discuss the different types of effluent treatment methods: Physical, chemical, biological and advanced technique (in brief only).	L	CO2	Understand	1
Water treatment methods	Draw the flow chart for the effluent treatment method.	L	CO2	Remember	0.5
Water treatment methods	Define potable water and list its characteristics.	L	CO2	Remember	0.5
Water treatment methods	Discuss the different steps involved in the municipal water treatment and explain each step briefly (Screening, sedimentation, coagulation, filtration and sterilization).	L	CO2	Understand	2
Water treatment methods	Explain different sterilization methods of water: Chlorination, UV radiation, and ozonation.	L	CO2	Understand	0.5
Water treatment methods	Draw the flow chart for municipal supply of potable water.	L	CO2	Remember	0.5
Water treatment methods	Explain different filtration methods of water (reverse osmosis and nano-filtration).	L	CO2	Understand	0.5
Acids ,bases - pH and POH	Define acids and bases according to the Arrhenius concept with examples. Define the terms pH and pOH. State the relation between pH and pOH. Define pH scale.	L	CO2	Understand	1
Acids ,bases - pH and POH	Solve problems based on pH.	L	CO2	Apply	1.5
Acids ,bases - pH and POH	Discuss the applications of pH.	L	CO2	Understand	0.5
Quantitative and Qualitative analysis	Standardisation of hydrochloric acid using standard sodium carbonate solution.	P	CO2	Apply	3
Quantitative and Qualitative analysis	Estimation of sodium hydroxide using standard hydrochloric acid.	P	CO2	Apply	3
Quantitative and Qualitative analysis	Estimation of potassium hydroxide solution by titrating against standard oxalic acid solution.	P	CO2	Apply	3
Quantitative and Qualitative analysis	Standardisation of EDTA using ZnSO ₄	P	CO2	Apply	3
Quantitative and Qualitative analysis	Volumetric estimation of total hardness of given water sample using standard EDTA solution.	P	CO2	Apply	3

Quantitative and Qualitative analysis	Determine the pH of solutions using various techniques- pH meter, Universal indicator and pH test paper.	P	CO2	Apply	3
Module III					
Organic compounds with some important functional groups	Define functional groups in organic Chemistry and explain briefly about organic compounds with some important functional groups (basic idea only): alcohol, acid, ester, hydrocarbons(aliphatic, aromatic, saturated and unsaturated- basic idea only)	L	CO3	Understand	2
Dyes	Define dyes and differentiate between chromophore and auxochrome with examples.	L	CO3	Understand	1
Dyes	Classify dyes based on source : Natural and synthetic dyes with one example and one use for each.	L	CO3	Understand	1
Dyes	Describe the classification of dyes based on method of application: Direct, acid, basic, disperse, mordant and vat dyes with one example and one use for each.	L	CO3	Understand	2
Dyes	Classify dyes based on chemical structure: Azo, anthraquinone, nitro, indigoid dyes with one example and one use for each.	L	CO3	Understand	2
Dyes	Discuss about various dye-fibre interactions: Physical adsorption, hydrogen bonding, ionic and covalent interactions (basic idea only)	L	CO3	Understand	2
Pigments	Define pigments in Chemistry and explain briefly about types of pigments (inorganic and organic only). list the uses of pigments.	L	CO3	Understand	1
Experiments on preparation of dyes, esterification , identification of functional groups and test for unsaturation, and aromaticity	Demonstrate the preparation of azo dye.	P	CO3	Apply	1.5
Experiments on preparation of dyes, esterification , identification of functional groups and test for unsaturation, and aromaticity	Demonstrate the esterification reaction.	P	CO3	Apply	1.5
Experiments on preparation of dyes, esterification , identification of functional groups and test for unsaturation, and aromaticity	Demonstrate identification tests for functional groups.	P	CO3	Apply	3
Experiments on preparation of dyes, esterification , identification of functional groups and test for unsaturation, and aromaticity	Demonstrate test for unsaturation(Use Baeyer's test) and also test for aromaticity(Baeyer's test and ignition test)	P	CO3	Apply	3
Module IV					
Alloys	Define an alloy with examples. List the purposes of alloying. List any two applications of alloys like stainless steel, brass, bronze and solder.	L	CO4	Remember	1
Polymers	Define monomers, polymers and polymerization with examples and also differentiate between homo and copolymers - with one example for each.	L	CO4	Understand	1
Polymers	Differentiate between thermoplastics and thermosetting plastics- with one example for each.	L	CO4	Understand	1
Polymers	Discuss the monomers and any two applications of common polymers - PE, PVC, Nylon-66 , Bakelite, PET, PU, PBT, PC, PMMA.	L	CO4	Understand	2
Polymers	Name the monomer of natural rubber and explain the process of vulcanization of rubber and also list the properties of vulcanized rubber.	L	CO4	Understand	1

Polymers	Define synthetic rubber and List the monomers of synthetic rubber—Buna-S and Buna-N	L	CO4	Remember	1
Nanomaterials	Define nanomaterials and nanotechnology. Discuss the nanomaterials: Carbon nanotubes, fullerenes, and graphene(basic idea only) List the applications of nanomaterials.	L	CO4	Understand	1
Metal Organic Frameworks (MOFs)	Define Metal-Organic Frameworks (MOFs) and list any three applications.	L	CO4	Remember	1
Biomaterials	Define biomaterials and list any three applications. Discuss briefly about biomaterials (metallic, ceramic, polymeric, and composite). Define biofuel with examples and discuss the applications of biofuel.	L	CO4	Understand	1
Green Chemistry	Discuss the principles of Green Chemistry and the importance and applications of green chemistry.	L	CO4	Understand	1
Experiments based on estimation of metals in alloys and preparation of polymers	Estimation of copper in brass.	P	CO4	Apply	3
Experiments based on estimation of metals in alloys and preparation of polymers	Estimation of zinc in brass.	P	CO4	Apply	3
Experiments based on estimation of metals in alloys and preparation of polymers	Estimation of iron in iron ore	P	CO4	Apply	3
Experiments based on estimation of metals in alloys and preparation of polymers	Preparation of: Urea-formaldehyde resin	P	CO4	Apply	3
Experiments based on estimation of metals in alloys and preparation of polymers	Preparation of Phenol-formaldehyde resin	P	CO1	Apply	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Prepare an assignment to find the atomic number and mass number of any 20 elements. Draw the structure of any 5 atoms and label the nucleus and fundamental particles. Find the atoms and molecules present in 5 substances around you. Prepare a simple model of atom using chart/ coloured paper/ seeds etc. Completion of record work and result analysis.	6
2	Write a note on concentration, methods of expressing concentration terms. Find the normality and molarity of different solutions(simple problems) . Find 5-10 solutions available at home and prepare a short report including: Name of solution, Solute, Solvent etc. Completion of record work and result analysis.	6
3	Write a report on the different thermodynamic processes and identify daily life examples in detail. Observe endothermic and exothermic processes at home and make a report. Completion of record work and result analysis.	6
4	Find any 4 spontaneous and non-spontaneous processes in everyday life. Record the principle and procedure for the preparation of standard solutions. Completion of record work and result analysis.	6
Module II		
5	Assignment on extra problems based on normality equation ($N_1V_1=N_2V_2$) Make a study report on choice of indicators- Selection of suitable indicators for different types of titrations. Identify and prepare simple natural indicators using household materials. Completion of record work and result analysis.	6
6	Collect water samples from different sources and identify pH. Identify soft water vs hard water through daily-life observation. Make a report on problems caused by hard water at home. Completion of record work and result analysis.	6
7	Identify different methods involved in modern water filters. Write a report on major stages in municipal water treatment and draw the flow chart. Do the sedimentation and filtration using muddy water. Completion of record work and result analysis.	6

8	Prepare a report on water purification methods in your locality. Prepare for an assignment on CO1 and CO2 Prepare for lab series - 1 exam Completion of record work and result analysis.	6
Module III		
9	Suggest the different tests to identify functional groups-alcohol, acid and ester. Prepare a report on organic acids present in different fruits and vegetables. Completion of record work and result analysis.	6
10	Prepare a report on different natural and synthetic dyes used in textile printing. Watch a video on dyeing clothes and make a report on it. Completion of record work and result analysis.	6
11	Identify the reaction involved in esterification Prepare a short note on different ester compounds that you find in your daily life.. Completion of record work and result analysis.	6
12	Prepare a report on classification of different types of pigments used in textile/printing. Completion of record work and result analysis.	6
Module IV		
13	Prepare an assignment on articles made of alloys used in daily life. Write a note on bioplastics/biodegradable polymers. Present a report of the polymeric materials which can be reused and recycled. Prepare a short video of major nanomaterials and its applications. Completion of record work and result analysis.	6
14	Find the materials used for biomedical implants. Prepare an assignment on biomaterials. Prepare a report on the applications of metal organic frameworks. Completion of record work and result analysis.	6
15	Identify the procedure for any green synthesis. Prepare an assignment on biofuels. Completion of record work and result analysis. Prepare for an assignment on CO3 and CO4 Prepare for lab series - 2 exam.	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Principles of Physical Chemistry	Puri, Sharma & Pathania	Vishal Publishing.Co.
2	Textbook of Physical Chemistry	P.L.Soni, O.P.Dharmarha , U.N.Dash	
3	Vogel's Textbook of Quantitative Chemical Analysis	Arthur Israel Vogel	Prentice Hall, 2000
4	Fundamentals of Analytical Chemistry	A Douglas, Skoog and Donald M. West, F.J. Holler	8thEdn., Harcourt College Publishers, 2004
5	The Essentials: Understanding Nanoscience and Nanotechnology;	Pradeep, T.	1st Edition ed.; McGraw-Hill Education: New York, 2007
6	Graphene Synthesis, Properties,Technology and Applications	Saritha Appukkutan,Jitha S. Jayan, Sethulakshmi A.S	Woodhead Publishing
7	Nanoscience and Technology	Muralidharan, V. S.; Subramania	New Delhi : Ane Books for ; Boca Raton : CRC Press
8	Practical Chemistry	A.O. Thomas	Scientific Book Centre, Cannanore / Kannur
9	Water and waste water engineering	Sudha Goel	Cambridge University Press
10	Color Chemistry: Syntheses, Properties, and Applications of Organic Dyes and Pigments"	Dr. Heinrich Zollinger	Verlag Helvetica Chimica Acta (Zürich)
11	Industrial Dyes: Chemistry, Properties, Applications"	Dr.Klaus Hunge	
12	Organic Chemistry	Jonathan Clayden, Nick Greeves, and Stuart Warren	
13	Textile Dyes and Pigments: A Green Chemistry Approach"	Pintu Pandit et al.	

REFERENCE			
Sl. No.	Title	Author	Publication
1	Physical Chemistry	Atkins	Oxford University Press (OUP)
2	Organic Chemistry.Vol.-I	I. L. Funar	Pearson Education
3	Textbook of Physical Chemistry	P.L.Soni, O.P.Dharmatha, Dash U N	Sulthan Chand & Sons

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in/courses/104106119	

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Conductivity meter	LED/LCD display range upto 1000mS/cm(230V)	1
2	pH meter	Digital/benchtop/portable/pen type	1
3	Bunsen burner		2
4	Digital weighing balance		1
5	Thermometer		1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	
Exam mark		15	50	40	40	50	50	60	
Converted to		5	10	10	10	10	10		
Marks	5	5	10	10		10		60	
Total	40							60	
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10

6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

Note:

Experiments shall be conducted such that all COs are attained.

Minimum of 2 experiments from each module shall be performed and total 10 experiments from all modules should be performed.